

VIKAS SHUKLA

Software Engineer ML • Machine Learning Systems • Full-Stack SWE

716-903-2501 | vikasofficial927@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | Bellevue, WA

Software ML Engineer with 7+ years of experience building scalable, high-performance distributed systems. Proven technical leader with a strong track record of delivering innovative solutions, leading cross-functional teams, and driving measurable business impact.

EXPERIENCE

Sr. Machine learning Engineer | Capital One | Richmond, VA May 2023 – Present

- Architected and deployed scalable real-time ML inference services supporting fraud decisioning across 40M plus transactions, improving model precision by 17%, reducing analyst effort by 45 hours/month, and enabling highly available, low-latency production predictions.
- Built platform-level observability framework for ML pipelines including automated data validation, KPI monitoring, alerting, and deployment health checks, reducing production incidents by 80%
- Designed and built a scalable data ingestion and transformation layer consolidating 40M+ transactions from disparate systems into a unified self-serve analytics platform, eliminating 60% of manual analysis cycles.
- Designed and deployed containerized ML microservices with CI/CD automation, model versioning, rollback capabilities, and zero-downtime production releases.
- 60% coding & implementation, 10% system design (HLD; LLD), 25% teams operational excellence(on-call) and 5% hiring/mentoring.

Software ML Engineer | Amazon | Bellevue, WA Jun 2022 – Apr 2023

- Designed and deployed secure machine learning development infrastructure supporting model training, experimentation, and deployment workflows for the Alexa Data Science organization.
- Built AI-powered notification quality assessment framework integrating user settings, reachability signals, and A/B test results across push, SMS, and email channels; improved Alexa user engagement and retention by 20%.
- Launched uplift-based targeting system with guardrails against notification fatigue: reduced total sends by 22% with no engagement loss, cut SMS/email failures by 30%, and lifted incremental conversion by 8%.
- Implemented automated business logic for A/B, multivariate, and auto-targeting marketing experiments using Python, increased campaign ROI and strategy effectiveness.

Software Engineer | SGS Teknics | Gurugram, India Nov 2014 – May 2018

- Automated on-premises server fleet scaling for SGS tax engine during peak holiday traffic, reducing deployment time from 3-4 hours per server to just 25 seconds.
- Led the infrastructure design and deployment of SGS tax processing platform, ensuring high availability, scalability, and reliable performance during peak sales periods.
- Mentored interns, organized innovation hackathons, and delivered technical training programs to strengthen team capability and engineering excellence.

KEY PROJECT

IntelliQuery Engine (Agentic AI Platform | LLM + RAG) [🔗](#) | Python, FastAPI, FAISS, React Native, SQL

- Built Retrieval-Augmented generation system using LLM embeddings and FAISS vector search enabling semantic queries across 10K+ document chunks with <800ms response time; reduced unsupported responses by 60%.
- Implemented agentic routing and context-aware retrieval workflows for finance/legal queries, improving accuracy by 45% and reducing manual validation effort by 50%.
- Deployed scalable FastAPI backend integrated with React Native application ([caseFit](#)) delivering production-grade low-latency AI responses.

EDUCATION

M.S Information Studies & AI | Trine University, Reston, VA Feb 2025

M.S. Data Science (GPA: 3.7 / 4.0) | University at Buffalo Feb 2022

B.Tech. Electrical & Electronics Engineering (GPA: 8.6 / 10) | SRM University, India May 2014

TECHNICAL SKILLS

Languages: Python, Java, C#, JavaScript, C++, SQL, React JS, React Native, TypeScript, FastAPI

Engineering: Object Oriented Programming, Scalable Architecture, Distributed Systems, Cloud Computing, Microservices
HLD, LLD, REST APIs, Data Structure and Algorithms, Linux

Cloud & DevOps: AWS Services (Lambda, S3, ECS Fargate, DynamoDB, RDS, SQS, SNS, Kinesis, CloudFormation, Glue Event Bridge, AppConfig), Azure Services (CosmosDB, Kafka, Kubernetes)

Generative AI: LLMs, RAG, Vector Databases, Agentic AI, Prompt Engineering, LangGraph, CrewAI, OpenAI APIs, AI Agents